Project Title
Threat-Intelligence Kill-Chain Correlation Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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Submitted by
K. Usharani (2520090049)
Samyuktha Kandukuri (2520090157)

Team Number: 07
Section: 11

Under the guidance of

Mrs. Ch. Anitha

Asst. Professor, CS&IT

Hyderabad-500090, Telangana, India.

ABSTRACT

Threat-Intelligence Kill-Chain Correlation Engine is a cybersecurity application developed **using Java and Data Structures & Algorithms (DSA)** to identify, analyse, and correlate cyber threats by reconstructing the sequence of an attack using the Cyber Kill Chain framework. Modern organizations generate a large volume of security logs and alerts, making it difficult to manually identify complete attack patterns. The proposed system analyses security log files collected from multiple sources and detects suspicious activities such as repeated failed login attempts, unauthorized access, privilege escalation, and abnormal data access. Related security events are correlated to reconstruct the attack timeline, allowing security analysts to understand the progression of an attack from initial access to its final objective. The system also identifies attack paths, highlights critical points that require immediate attention, prioritizes threats based on their severity, and provides a centralized dashboard for monitoring and analysis. By automating threat correlation and reducing manual effort, the proposed solution improves threat visibility, enhances incident response, minimizes false positives, and strengthens an organization's overall cybersecurity posture through efficient and intelligent threat analysis.

Signature of the Guide